

Foundation-model morphology maps residual programs in spatial-omics contours

Abstract

New whole-transcriptome in situ platforms measure >18,000 genes in FFPE tissue, creating a need to test whether co-registered H&E captures residual molecular programs. We introduce morphomolecular translation mapping (mTM), which tests whether H&E foundation-model embeddings map residual WTA programs after contour labels and spatial covariates are fixed. In breast and cervical Atera WTA data, mTM nominates endocrine, stress, stromal, immune and boundary-associated programs exceeding spatial nulls.

Spatial biology is moving from sparse assays to high-resolution transcriptome maps that preserve tissue context¹⁻⁵. Atera whole-transcriptome in situ profiling recently extended this direction to FFPE tissue-scale measurements of more than 18,000 genes, including public breast and cervical cancer preview data with 18,028 profiled genes²⁰. This creates a new integration problem: transcriptome-scale spatial assays now define rich molecular contours, but the residual molecular variation within those contours may still be visible to co-registered H&E morphology. These data are not simply a larger targeted panel; they make it possible to ask, in one registered FFPE tissue system, whether endocrine, stress, stromal, immune and boundary-associated WTA programs retain morphologic structure after the assay itself has defined molecular-spatial regions. In parallel, H&E foundation models and histology-spatial-omics systems⁶⁻¹⁶ now support visual-language pathology, whole-slide representation learning and expression-morphology integration. Most work asks whether H&E can reconstruct expression, spatial domains or downstream labels. We asked a narrower question: after spatial omics has already defined molecular-spatial tissue contours, do H&E embeddings still map the WTA variation left unresolved by those contours? This shifts the benchmark to residual biological decoding and

reduces the risk that an H&E-WTA association merely reflects region identity, slide geography or boundary structure.

We developed morphomolecular translation mapping (mTM), a contour-constrained residual decoding framework (Fig. 1) adapted to these newly available WTA-scale spatial datasets. Spatial-omics-derived contours were generated from Atera WTA cell-coordinate and cluster information, not from H&E image segmentation. The same polygons served as shared geometry^{18,19} for direct-WSI LazySlide PLIP or UNI H&E embeddings^{12,14,17}, WTA program summaries and residual tests. The contours are therefore not independent histological labels. The conservative estimand is whether H&E embeddings map continuous WTA variation within a spatial-omics label after centroid position and boundary-distance covariates are controlled. This design avoids a weaker interpretation in which morphology appears successful because it only detects tumor, stroma, necrosis or slide geography already encoded by WTA clustering. By residualizing within the molecular-spatial label and preserving local spatial bins during permutations, the analysis targets continuous program differences among contours placed in the same broad molecular state. The output is a ranked set of contour-level program-feature pairs, effect sizes and spatial-null checks, rather than a diagnostic classifier. This contour scale is larger than a tile, retains local context and keeps permutation/bootstrap units aligned with biologically meaningful regions.

In breast WTA, the strongest evidence came from S3 contours, which define a molecular-spatial class but are not molecularly homogeneous. mTM places each S3 contour along residual continua of endocrine, stress and metabolic state. Within S3, PLIP-derived morphology was associated with residual luminal estrogen-response variation (partial Spearman's rho -0.639, $P = 4.79 \times 10^{-19}$, $n = 157$), unfolded-protein-response variation (partial Spearman's rho 0.515, $P =$

7.91 $\times 10^{-12}$) and oxidative-phosphorylation variation (partial Spearman's rho 0.531, $P = 1.38 \times 10^{-12}$). These are contour-level residual associations, not per-cell or per-tile measurements. Because image-axis direction is arbitrary, the result means contours sharing the same spatial-omics label retained continuous endocrine, stress and metabolic WTA variation mapped by H&E features. The luminal estrogen-response result is the clearest discovery example because it occurs after restricting to one discrete contour label and after spatial covariate adjustment; it was not driven by very small contours, because requiring $n_tiles \geq 3$ and $n_cells \geq 50$ gave a recomputed partial Spearman's rho of -0.656 ($n = 49$). Hero patches in Fig. 1b are therefore QC-filtered representative examples selected after statistical ranking, with expanded examples and metadata in Supplementary Fig. 2 and Supplementary Table 4, not a blinded visual diagnostic claim.

We next tested whether these associations were simply consequences of spatial autocorrelation. Molecular residuals were permuted within strata preserving spatial-omics-derived contour label, centroid-position bins and boundary-distance bins. The breast luminal estrogen-response, unfolded-protein-response and oxidative-phosphorylation associations remained stronger than the compartment-aware null distributions, each with empirical $P = 9.999 \times 10^{-5}$ (Supplementary Fig. 1 and Supplementary Table 1); block-bootstrap intervals are reported in Supplementary Table 2. Across breast and cervical analyses, 13 candidate associations passed the 95% spatial-null gate, and the strongest candidates also exceeded the 99% null threshold. Additional sensitivity checks retained candidate signs under leave-one-block, local-mismatch, centroid-jitter and 23 small tile-registration perturbation tests; nested spatial holdout recovered the primary breast PLIP signs in 75.0-93.8% of folds, with spatial-sensitivity, component-gene, registration-perturbation, nested-holdout and contour-size summaries in Supplementary Tables 5-9. A

cervical Atera WTA dataset served as a cross-cancer stress test rather than a direct replication of the breast luminal phenotype. It was used to ask whether the same residual-decoding procedure could nominate plausible program families in a different epithelial cancer, not whether breast-specific luminal biology would recur. In that second epithelial cancer context, residual decoding recovered stromal-remodeling, immune-ecology and invasion-associated programs in 215 paired H&E-WTA contours, including myofibroblast CAF activation, collagen/ECM organization and stromal encapsulation; TLS-adjacent activation, immune exclusion and immune activation; and EMT/invasive-front and EMT-invasion programs. Epithelial-identity and oxidative-phosphorylation candidates provided additional epithelial and stress/metabolic support. PLIP and UNI agreement was evaluated only at the program-family level, because embedding axes can rotate or change sign between models, with program-family summaries in Supplementary Table 3. Thus, the cervical result is consistent with applying the residual-decoding workflow to a different epithelial cancer setting, while cohort-level generality remains a future test.

Finally, ring-level boundary profiles nominated candidate molecularly active zones, defined here as interfaces where H&E and WTA gradients co-varied at selected tissue boundaries (Supplementary Fig. 3). The current data support boundary co-variation, not causality, temporal ordering or a directional mechanism. Atera WTA provides an unusually broad 18,028-gene in situ discovery layer²⁰, but transcript evidence does not replace protein validation. For this reason, the analysis treats Atera WTA as the measurement layer defining the biological coordinate system, and H&E as an independent modality asked only to explain residual structure. mTM therefore nominates morphology-associated molecular states and boundary ecologies for validation in larger cohorts and by IHC or orthogonal molecular assays where clinically relevant. Its immediate contribution is a residual test showing that foundation-model morphology can add

information within spatial-omics-defined contours rather than merely reproduce those contours. More broadly, mTM reframes H&E-spatial-omics integration as a controlled interrogation of what remains after molecular-spatial labels are fixed, a question made newly practical by transcriptome-scale spatial platforms.

Data availability

Figure source-data CSVs for Fig. 1 and Supplementary Figs. 1-3 are included with the submission package. Separate machine-readable Supplementary Data CSVs are provided for robustness tables, including contour-size sensitivity, that are too detailed to rely only on PDF rendering. Raw 10x Genomics Atera WTA and H&E input files are not redistributed with this submission package; the breast and cervical source datasets are available from 10x Genomics at <https://www.10xgenomics.com/datasets/atera-wta-ffpe-human-breast-cancer> and <https://www.10xgenomics.com/datasets/atera-wta-ffpe-human-cervical-cancer>.

Code availability

The mTM analysis, direct WSI embedding, contour aggregation, statistical defense, figure composition and submission-package preparation code is available in the pyXenium repository at <https://github.com/hutaobo/pyXenium>. The fixed software release used for this submission is pyXenium v0.4.6 at <https://github.com/hutaobo/pyXenium/releases/tag/v0.4.6>, and the submission-packaging hardening commit is 3c4617010bb1c76e18921958c4507f91512aa9dc at <https://github.com/hutaobo/pyXenium/commit/3c4617010bb1c76e18921958c4507f91512aa9dc>. Manuscript-level replication scripts are provided under `manuscript/mtm_wta_nbt_replication/`. The exact journal-upload package and individual submission files are archived at

<https://github.com/hutaobo/pyXenium/releases/tag/nbt-submission-upload-20260518-draft>.
HistoSeg contour-generation software is maintained at <https://github.com/hutaobo/HistoSeg>.

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Figure legend

Fig. 1 | Contour-constrained residual decoding maps WTA programs from H&E morphology. a, Atera WTA-derived spatial-omics contours provide shared geometry for H&E LazySlide PLIP/UNI embeddings, WTA program summaries and residual tests. b, QC-filtered breast S3 luminal estrogen-response examples under the same contour label; labels report WTA and oriented H&E embedding z-scores. Displayed examples satisfy $n_tiles \geq 3$ and $n_cells \geq 50$, and are illustrative rather than diagnostic claims. c, Top residual programs exceed compartment-aware spatial-null thresholds. d, Candidate molecularly active zones show ring-level H&E-WTA boundary co-variation. e, Cross-cancer program-family stress test across breast/cervical PLIP/UNI candidates; dot area reports maximum absolute partial Spearman's rho and text reports the top WTA program. Agreement is interpreted at program-family level because embedding axes can rotate or change sign.

Declarations

Competing interests

The authors declare no competing interests.

158 **Ethics and data-use statement**

159 This computational analysis used public/vendor example spatial WTA and H&E data. The study
160 did not involve new human sample collection, intervention or generation of new identifiable
161 participant data.